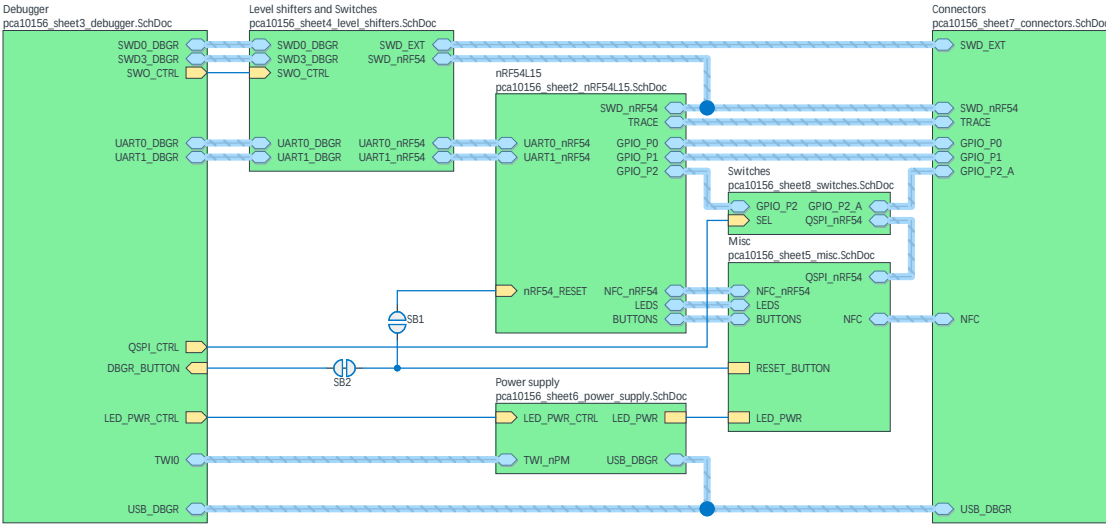


Nordic Semiconductor ASA

nRF54L15 DK (PCA10156)

- Sheet 1: Connections
- Sheet 2: nRF54L15
- Sheet 3: Debugger
- Sheet 4: Level Shifters
- Sheet 5: Misc
- Sheet 6: Power Supply
- Sheet 7: Connectors
- Sheet 8: Switches



✗ The No ERC object is a design directive.
This directive is placed on a node in the circuit to suppress harmless warnings and/or error r
violation conditions that are detected when the schematic project is compiled.

Logo's and
markings

PCA10156
1.0.0
<year> - <week>
<UID>



Board fiducials



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implementation. Nordic Semiconductor also does not claim that the design is
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your design implementation to confirm the functionality of your application.
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Title
nRF54L15 DK - Connections

Size PCB Assembly Number

A3 PCA10156

Date: 26.03.2025

File: pca10156_sheet1_connections.SchDoc

Classification: PUBLIC

Revision

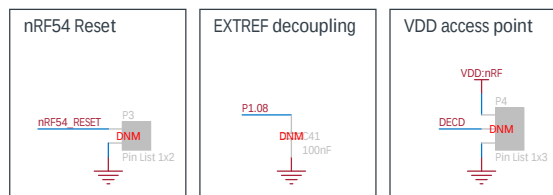
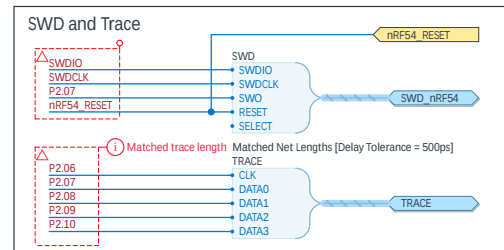
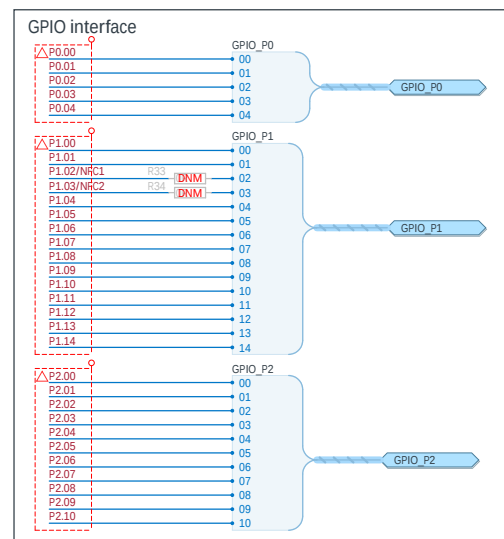
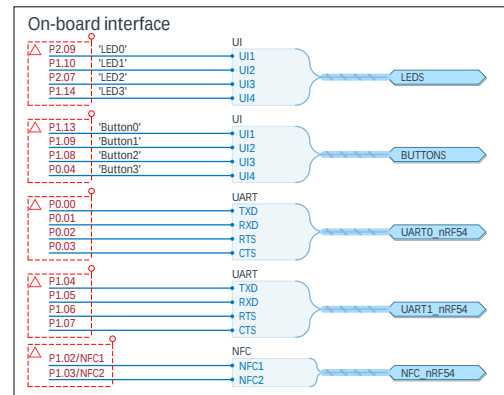
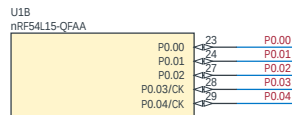
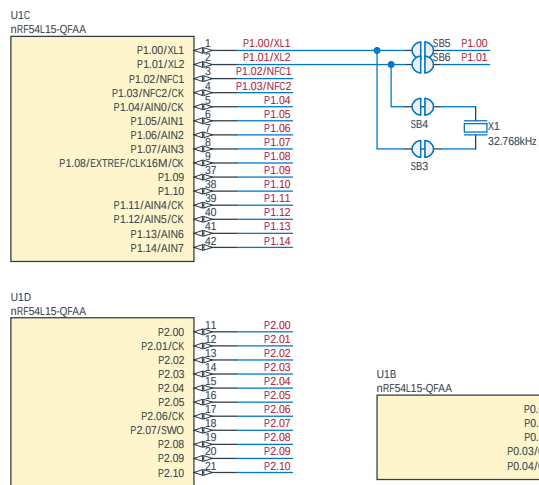
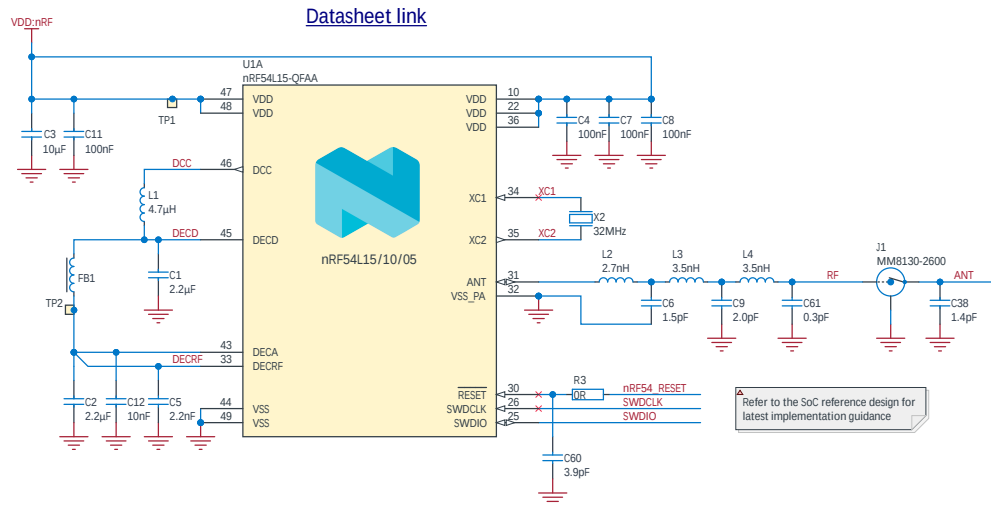
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Sheet 1 of 8

Drawn By: STL1



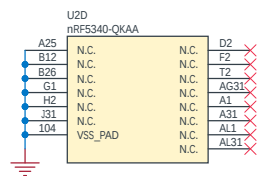
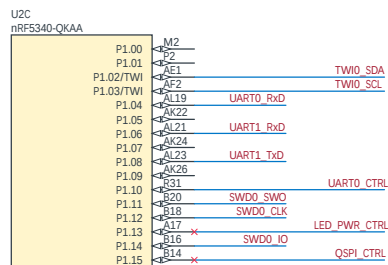
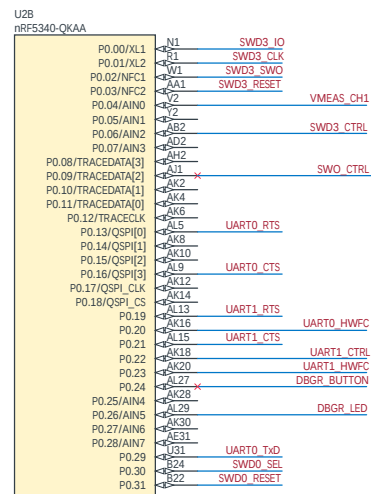
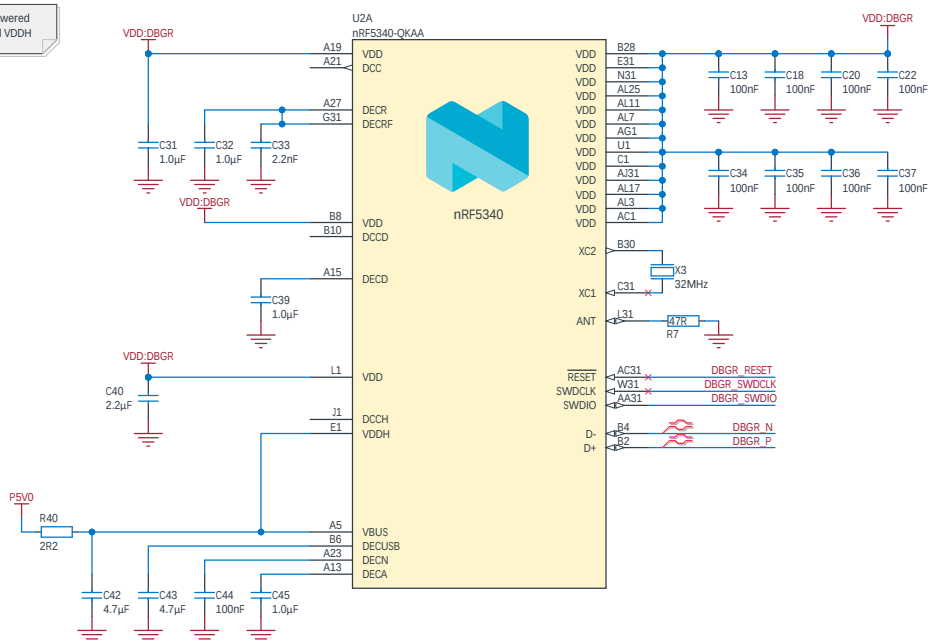
NORDIC
SEMICONDUCTOR



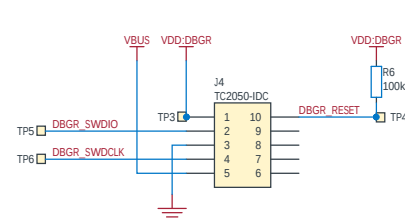
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Size A3	PCB Assembly Number PCA10156	Revision 1.0.0	
Date: 26.03.2025		Sheet 2 of 8	
File: pca10156_sheet2_nRF54L15.SchDoc		Drawn By: STL1	
Classification: PUBLIC			

Debugger

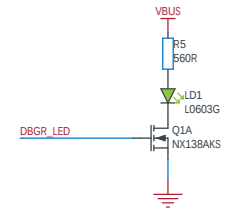
nRF5340 is self powered through VBUS and VDDH



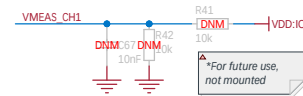
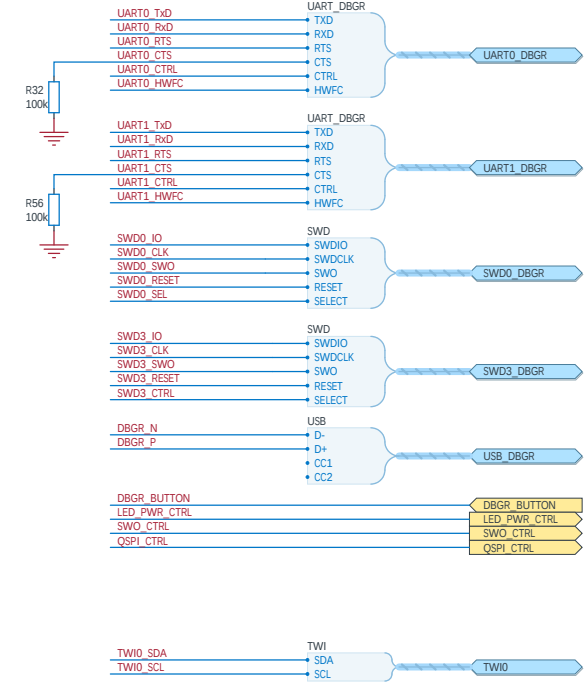
Debugger Programming Connector



LED



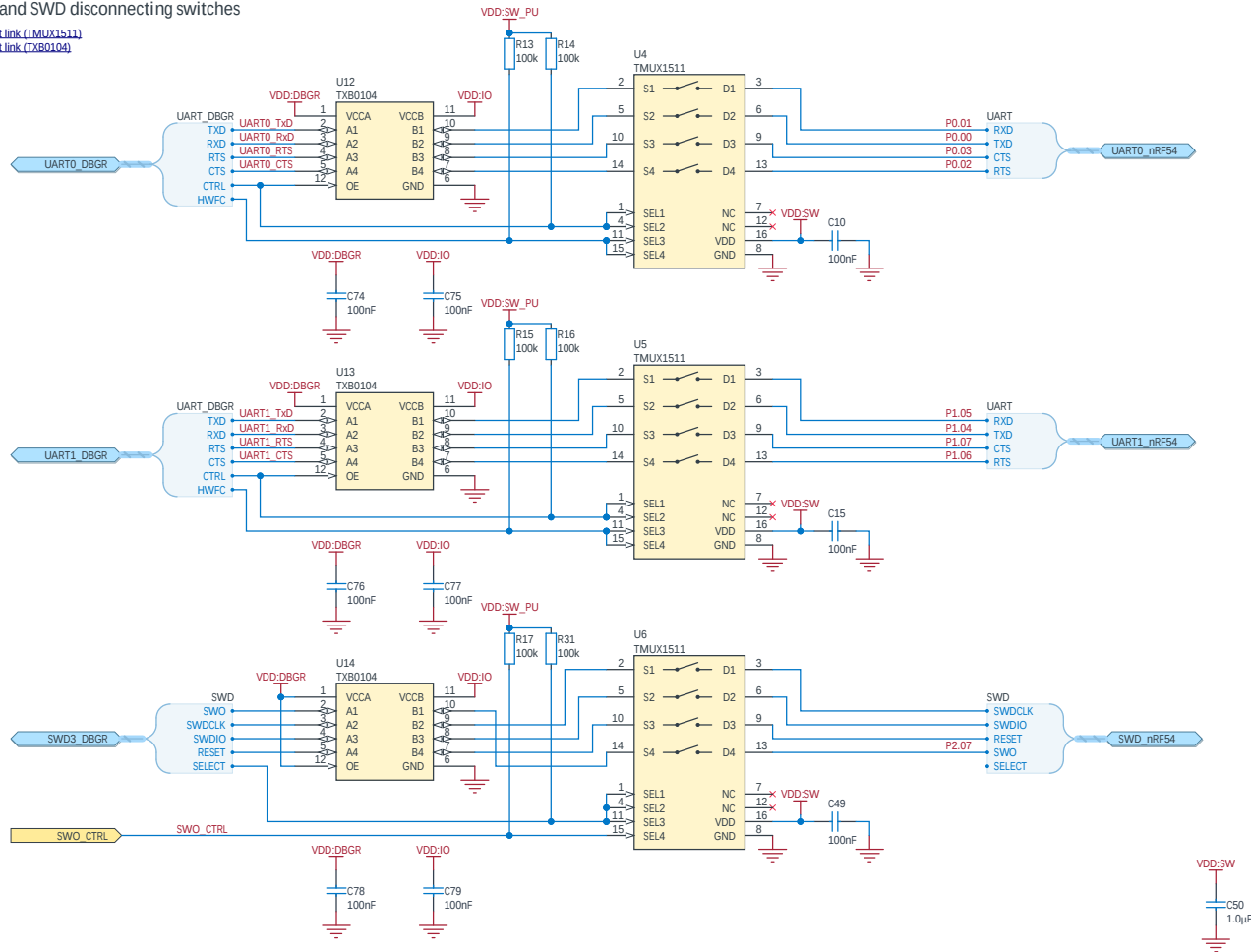
Ports



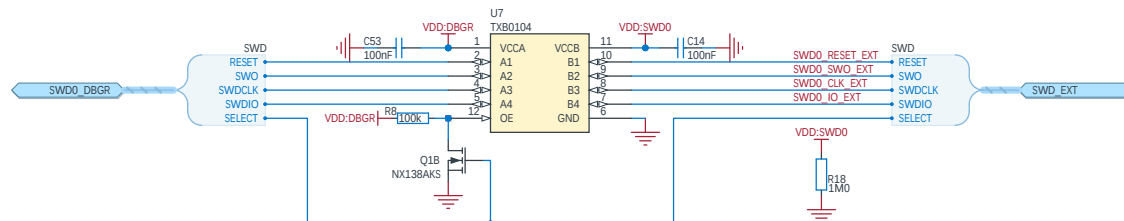
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Size A3	PCB Assembly Number PCA10156	Revision 1.0.0	
Date: 25.03.2025		Sheet 3 of 8	
File: pca10156_sheet3_debugger.SchDoc		Drawn By: STL1	
Classification: PUBLIC			

UART and SWD disconnecting switches

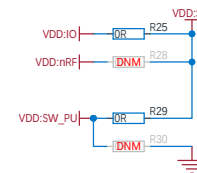
Datasheet link (TMUX1511)
Datasheet link (TXB0104)



SWD Debug Out level shifter



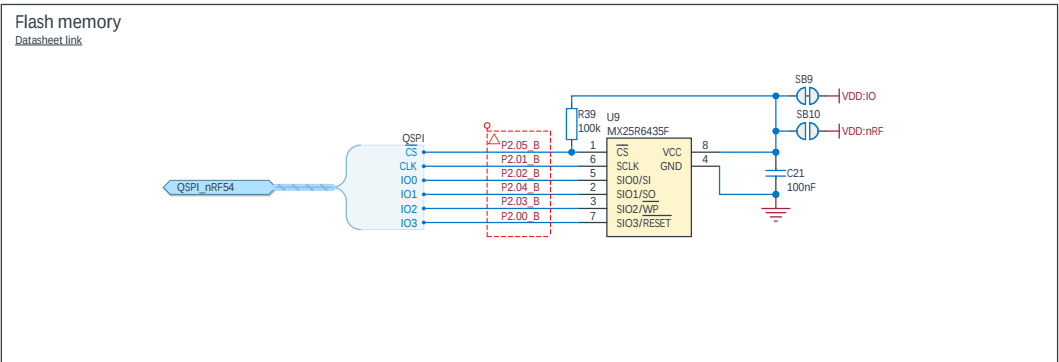
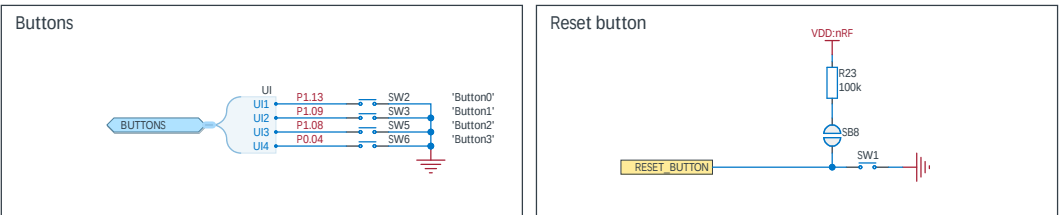
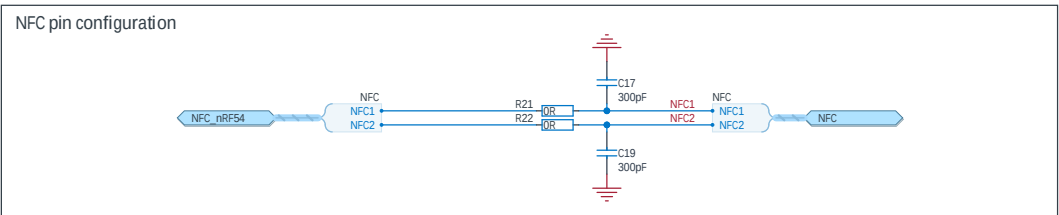
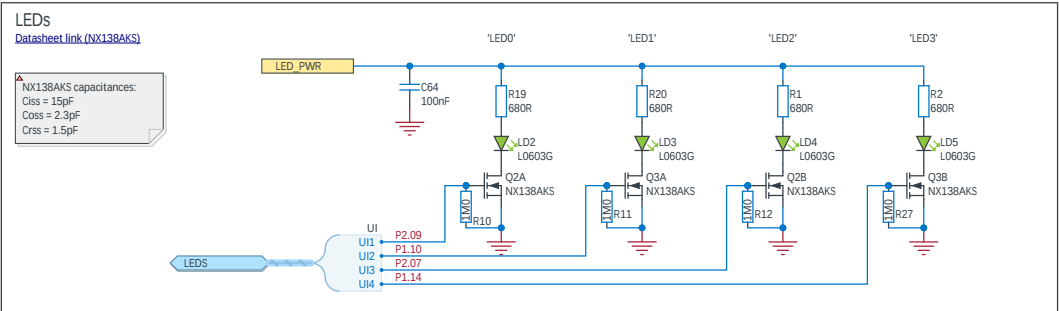
Analog switches manual override



Manual override for powering U1, nRF54L15, as standalone through P6 header.
R25, R29 mounted: Default configuration. DK requires power from USB connector.
R28, R30 mounted: Analog switches are powered from P6. Analog switches U3, U4, U5, U6 and U9 disconnects all signals. No external power needed from USB connector

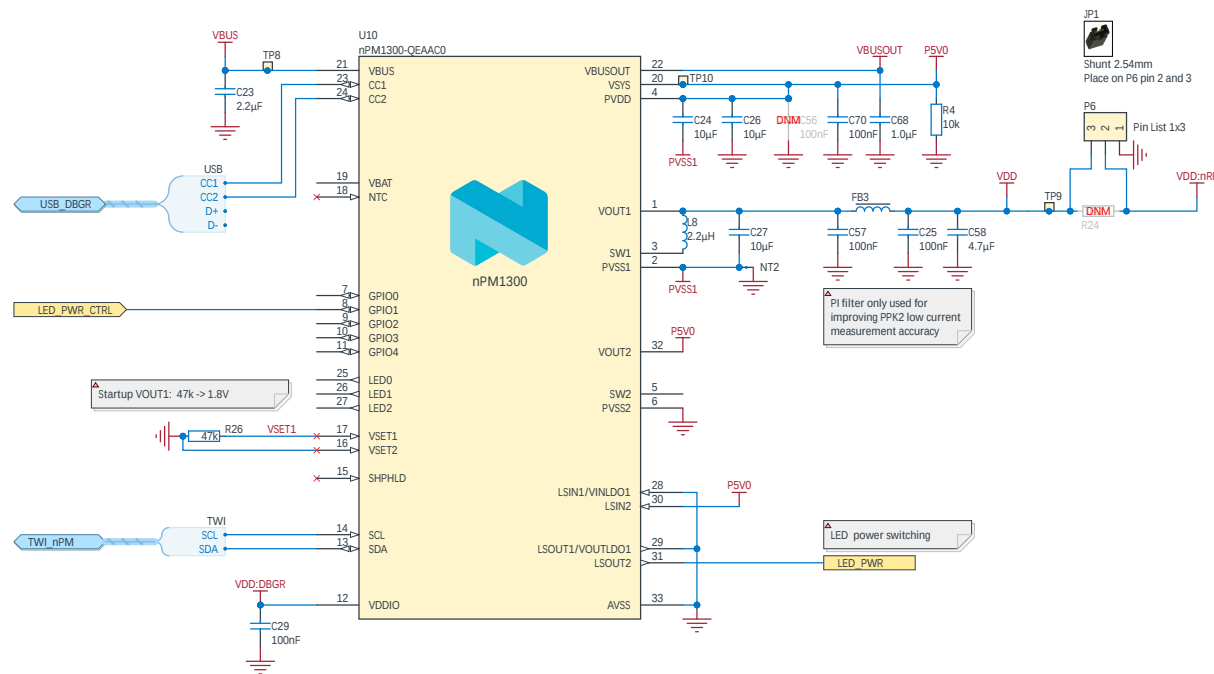
Title nRF54L15 DK - Level Shifters		Revision 1.0.0	
Size A3	PCB Assembly Number PCA10156	Sheet 4 of 8	
Date: 26.03.2025		Drawn By: STL1	
File: pcal0156_sheet4_level_shifters.SchDoc		Classification: PUBLIC	





Power managment

[Datasheet link](#)



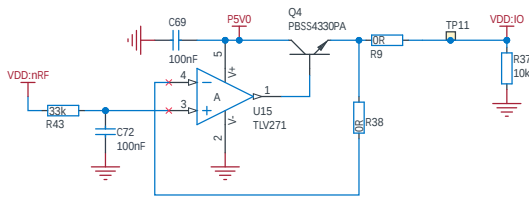
Power domain overview:

VDD	1.8V - 3.3V VDD source from nPM1300 VOUT1
VDD:nRF	Main VDD for nRF54L15, 1.8V to 3.3V
VDD:IO	Buffered VDD:nRF supplying analog switches, levelshifters, external memory
VDD:DBG	Debugger VDD (self powered)
VDD:SW_PU	VDD connection for all pullup resistors
VBUS	USB VBUS after power switch
VBUSOUT	VBUSOUT from nPM1300. Not used
PSV0	5V VSS (VBUS) used for input to Buck regulators and system 5V

nPM1300 settings:

- VOUT1 set to 1.8V
- VOUT1 enabled and forced in PWM mode
- VOUT1 output discharge enable
- Enable LS2 for LED power switching, using GPIO1 as enable/disable signal.

VDD:nRF supply buffer

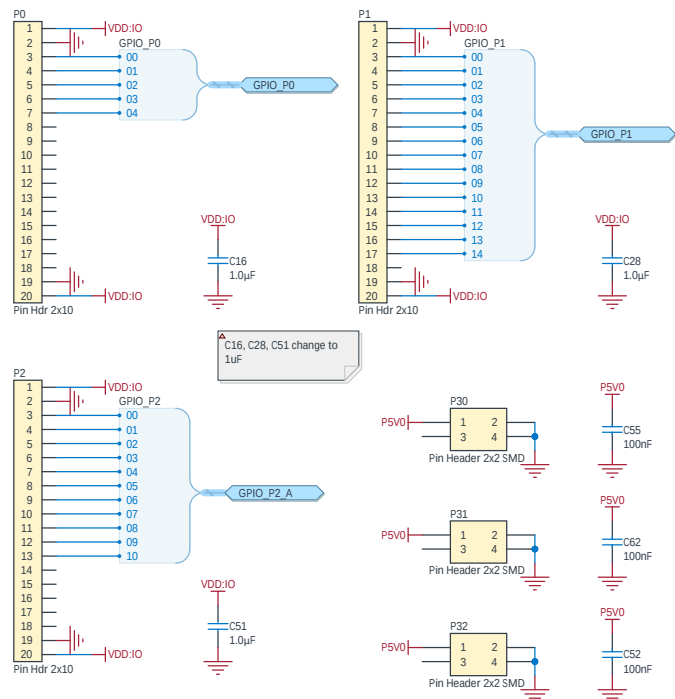


The VDD:nRF buffer is not needed for nRF54L15 reference design. Only used for DK operation and peripherals

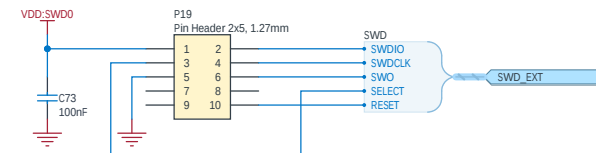


Title nRF54L15 DK - Power Supply		Revision 1.0.0	
Size A3	PCB Assembly Number PCA10156		
Date: 26.03.2025	File: pca10156_sheet6_power_supply.SchDoc	Sheet 6 of 8	Drawn By: STL1
Classification: PUBLIC			

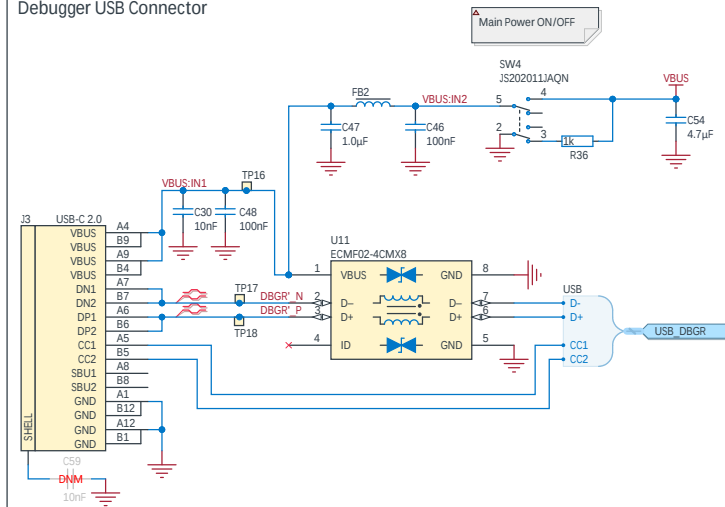
GPIO Connectors



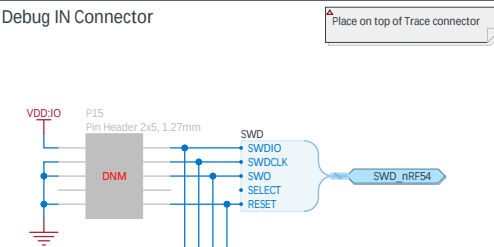
Debug OUT Connector



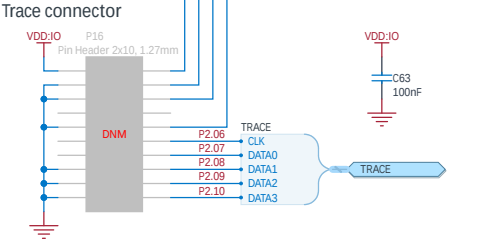
Debugger USB Connector



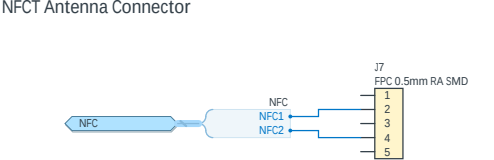
Debug IN Connector



Trace connector



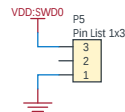
NFCT Antenna Connector



GND probe point

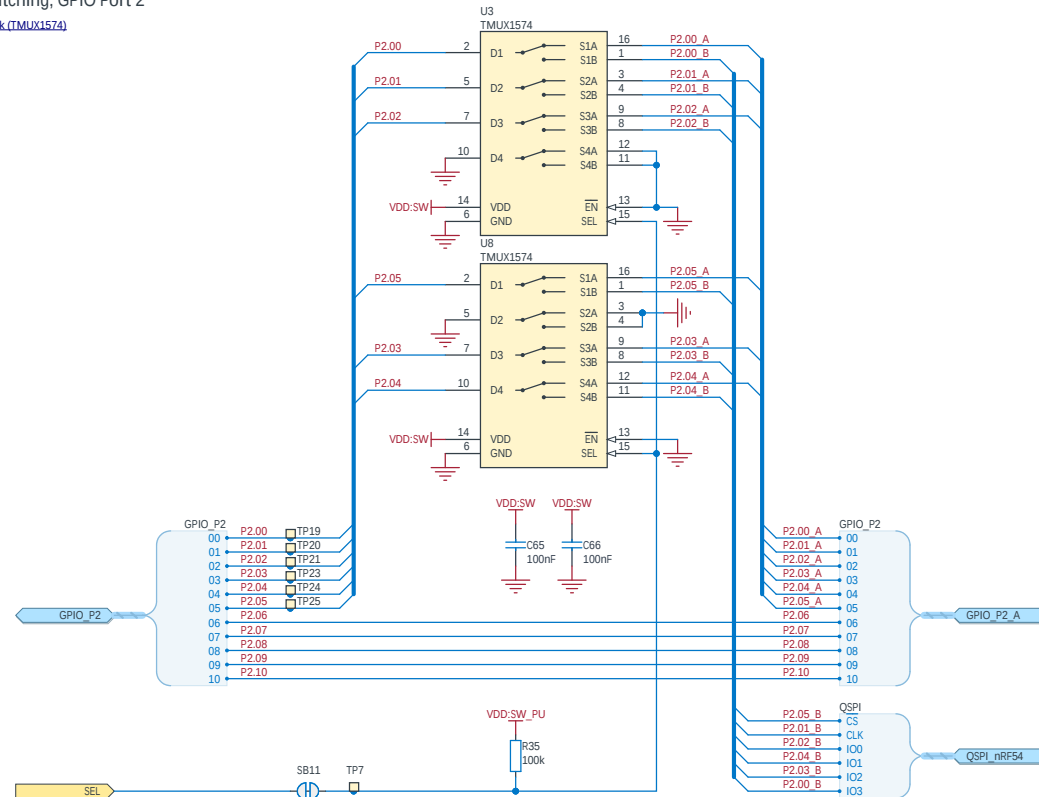


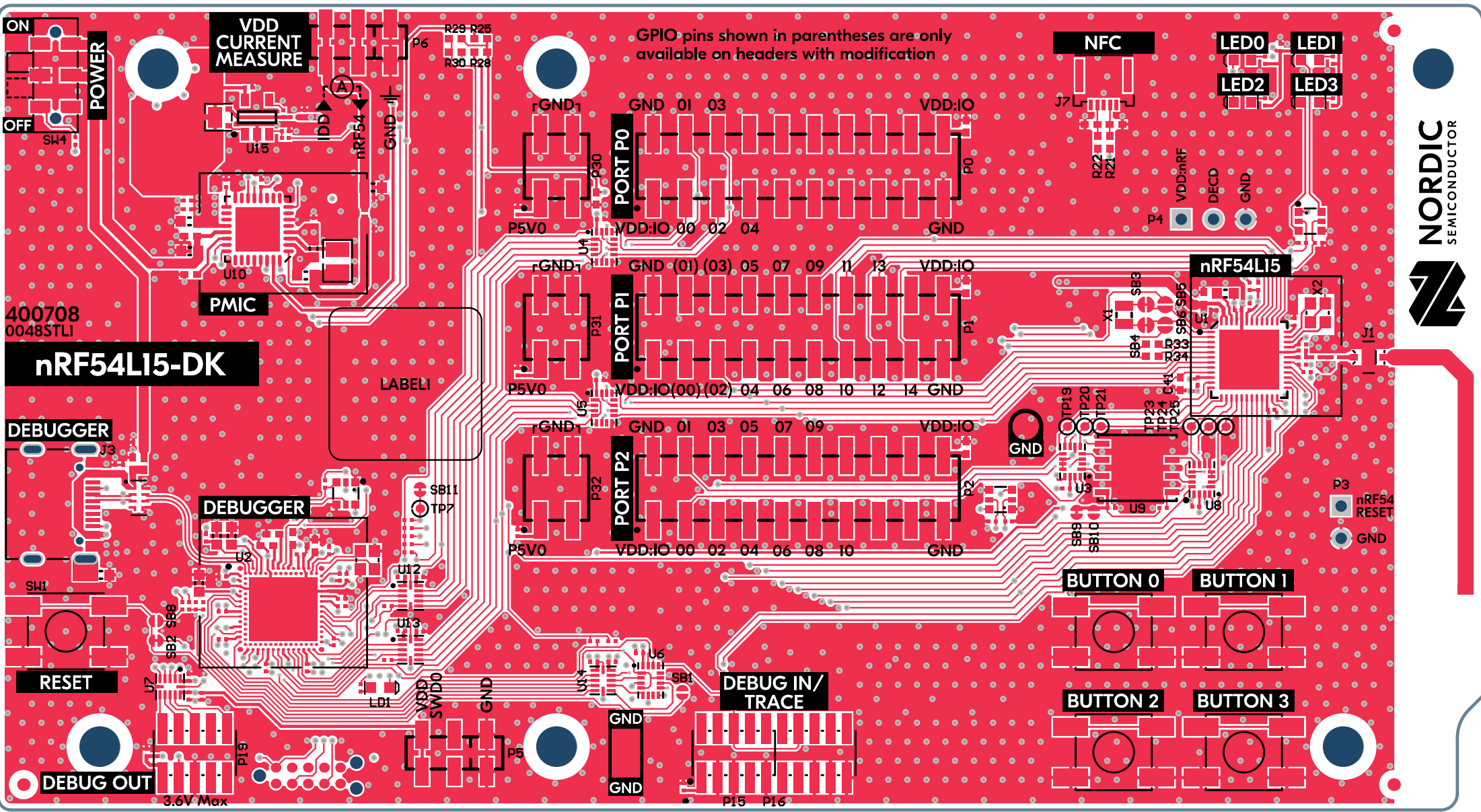
Power Connector

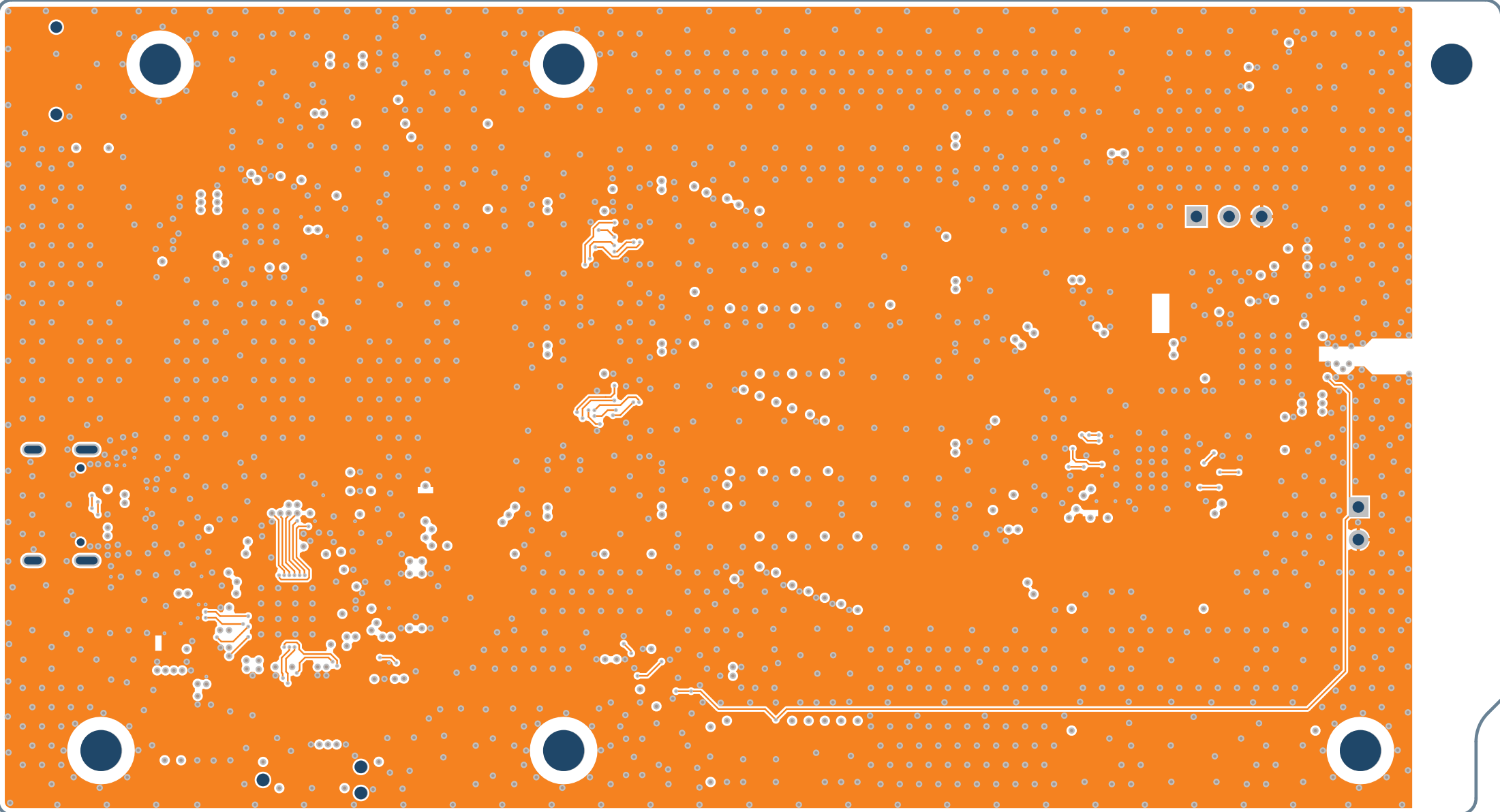


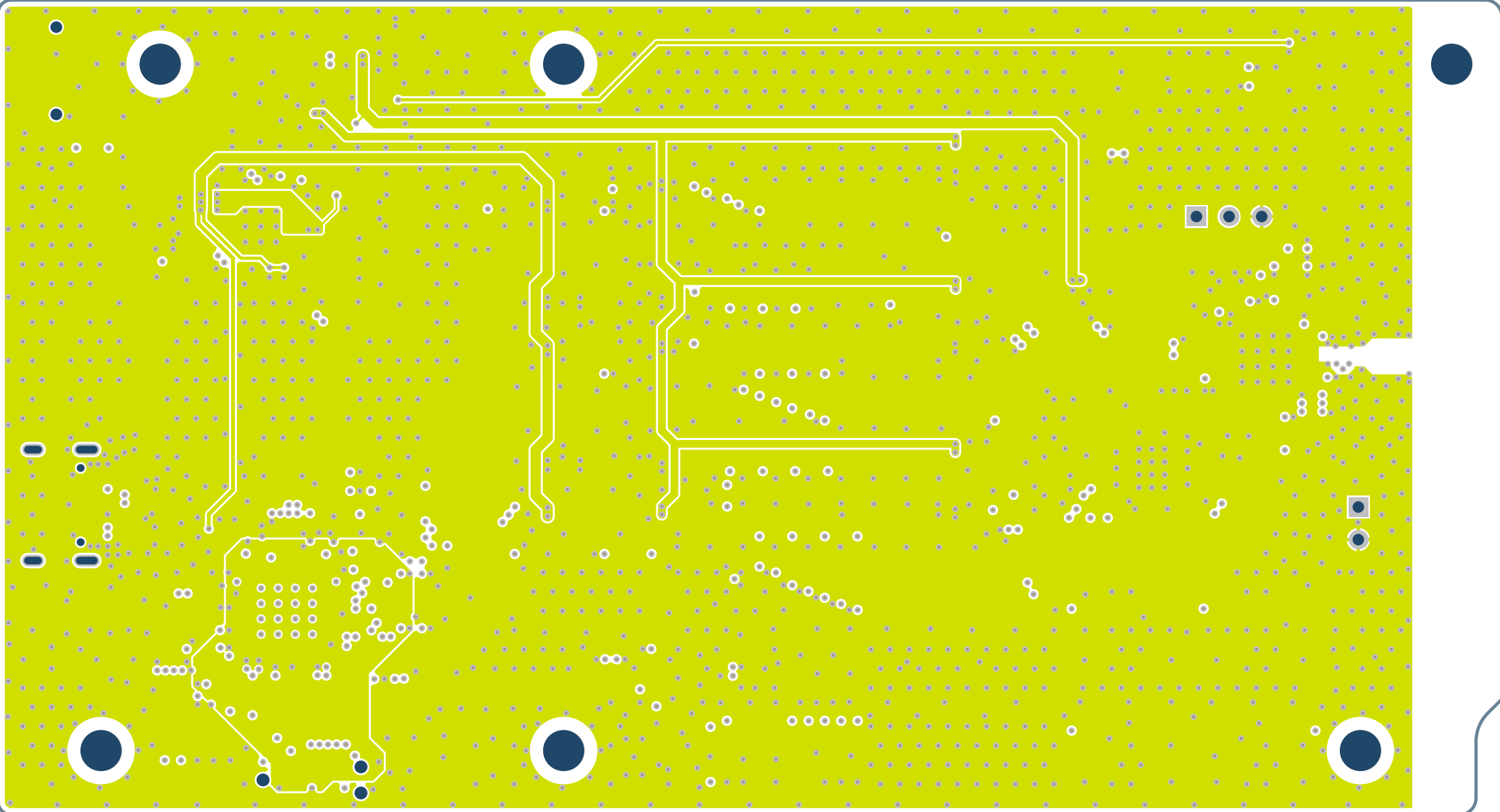
QSPI switching, GPIO Port 2

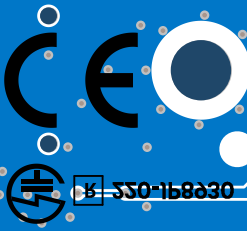
[Datasheet link \(TMUX1574\)](#)









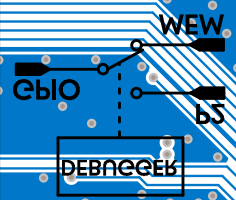


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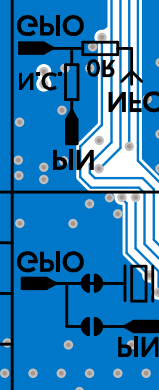


B5'03	ZMO	УВЕЛ24
	ZMDCIK	DEBUCEEB
	ZMDIO	
	BE2EL	
B1'03	CT2I	УВЕЛ24
B1'09	BT2I	
B1'04	LXD1	
B1'02	BXD1	БИИ
B0'03	CT20	DEBUCEEB
B0'05	BT20	
B0'00	LXD0	
B0'01	BXD0	
GBIO	Function	Circuit

B5'00	DIO3
B5'03	DIO5
B5'04	DIO1
B5'05	DIO0
B5'01	CFK
B5'02	C2
GBIO	Function



B1'03	IECS	GBIO	ИЕС
B1'03	IECI		
B1'01	XG3		
B1'00	XG1	2B2	2B4
		2B2	2B3
GBIO	Function	2port	NM



B0'04	BUTTON 3	
B1'08	BUTTON 5	
B1'06	BUTTON 1	
B1'13	BUTTON 0	
B1'14	GED 3	
B5'03	GED 5	
B1'10	GED 1	
B5'06	GED 0	
GBIO	Function	NM

